AI for Product Development

Activity 03: Build a Research Evidence Pack

C840 • v1.0

Transform mock market and competitor records into a traceable, uncertainty-aware insight brief.

1. Purpose and deliverable

Transform mock market and competitor records into a traceable, uncertainty-aware insight brief.

Required output: Claim ledger, competitor matrix and three prioritised insights

Acceptance: Material insights cite at least two records or explicitly state single-source uncertainty.

2. Folder inventory

brief.yaml — machine-readable scope and decision contract

prompt.md — bounded agent instruction

data/mock-data.csv and mock-context.json — synthetic evidence

solution/expected-output.json — trainer reference shape

scripts/verify_activity.py — deterministic completeness check

evidence/checklist.md — learner sign-off record

3. Detailed procedure

1. Read brief.yaml and restate the product decision, output and acceptance rule.

2. Inspect the CSV and JSON inputs; list source IDs, dates, contradictions and missing evidence.

3. Open prompt.md and select an approved agentic AI builder. Do not add live data or integrations.

4. Ask the agent to plan the workflow before producing the artifact. Check roles, tools, data and stop conditions.

5. Execute the activity-specific workflow: Load records → Normalise facts → Triangulate claims → Flag conflict → Rank insight → Cite sources.

6. Require citations to source IDs and label every inference or assumption.

7. Review the output at the human gate; compare it with the acceptance rule and reject unsupported claims.

8. Save the artifact and evidence inside a learner working copy of this folder.

9. Run python3 scripts/verify_activity.py and retain the JSON result.

10. Complete evidence/checklist.md and record PASS, REPAIR or STOP with a named owner.

4. Troubleshooting

Missing citations: require row-level source IDs and rerun only the unsupported section.

Unstable recommendation: freeze criteria, rerun the same input and perform sensitivity analysis.

Verifier failure: compare the missing file or field with brief.yaml, repair it and rerun.

Unsafe action proposed: stop, remove live/external access and return the decision to the human owner.

5. Verification

Pass condition: Material insights cite at least two records or explicitly state single-source uncertainty.

The deterministic verifier checks package completeness; the human reviewer checks evidence quality and decision fitness.